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قطر للطاقة
QatarEnergy

# IS PROCEDURE FOR CONTROL OF ASBESTOS

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## 1.0 PURPOSE

The purpose of this Procedure is to outline responsibilities and measures necessary to prevent or minimize the exposure of personnel when working with or encountering asbestos containing material (ACM).

## 2.0 SCOPE AND APPLICABILITY

This Procedure describe the primary precautions that shall be taken by all personnel when working with or encountering potential asbestos containing material and apply to all Idd El-Shargi worksites.

## 3.0 ROLES AND RESPONSIBILITIES

QatarEnergy-IS shall apply best industry practice to protect employees who should be exposed to Asbestos at work and manage any known installations of asbestos in their buildings or work sites. QatarEnergy-IS shall also protect other persons affected or at risk of being affected by potential exposure

### 3.1 OFFSHORE INSTALLATION SUPERINTENDENT (OIS)

Shall have overall responsibility for ensuring Best Industry Practice is maintained and this Procedure adhered for work involving ACM.

### 3.2 SAFETY OFFICER

- Maintaining and update a register, which indicates the known location of Asbestos on the worksite.
- Inspect the integrity of ACM.
- Marking the location of known Asbestos with warning labels, approved in advance by the HSE Department.
- Control on-site supervision of asbestos removal contractors and independent occupational hygiene services.
- Control the removal of samples and ensure these are dispatched to the relevant person onshore for analysis.
- Where applicable maintain a record of onsite clearance test certificates.

### 3.3 WORKSITE SUPERVISORS

- Liaise with the onsite Safety Representative before carrying out Minor Works on materials known to contain Asbestos, or in areas adjacent to known Asbestos locations where the potential of fiber release exists.
- Report to the Safety Representative the discovery of material thought to be ACM.
- Ensure that Permit to work system is followed for activities related to ACM.
- Ensure that Area containing ACM during ACM related activity i.e maintenance, sampling & removal access shall be restricted and barricaded.

### 3.4 SUPPLY CHAIN

Shall ensure that in accordance with Best Industry Practice any Asbestos waste is correctly packaged, labelled and documented for transportation onshore and ultimate disposal in accordance local laws and regulation

## 4.0 PROCEDURAL STEPS

### 4.1 REQUIREMENTS

**Work Method**

### 4.1.1 GENERAL

The introduction of ACM is prohibited for all applications on all QatarEnergy-IS operated worksites. All work with ACM, whether it is major or minor (as defined in section 3.0), can potentially lead to fibre release. It is therefore vital that HSE are consulted in advance for all work, which shall be planned, and risk assessed. Appropriate precautions shall be adopted, prior to the commencement of any work involving handling of ACM.

#### 4.1.1.1 Where Asbestos may be found

Despite its well-recognized dangers, asbestos has many useful properties, which led to its widespread use, particularly for fire and thermal insulation and as a binding agent within other materials, therefore in older buildings and work sites there is always a possibility of ACM being found in existing structures or applications. Asbestos products may be encountered in the following situations:

- Asbestos lagging
- Asbestos insulation board or blocks
- Asbestos cement products
- Sprayed asbestos coatings
- Asbestos tiles and felts
- CAF gaskets

#### 4.1.1.2 Identification of Asbestos

Where a material is encountered and it is suspected that it may contain asbestos, a positive identification shall be made before further work is undertaken.

The identification may be carried out by:

- Representative sampling (bulk) and analysis
- Reference to plans or records
- Use of information from suppliers.

Any material specified as calcium silicate and used in the construction of boilers, fireboxes, furnaces etc. shall be regarded as suspect and should not be disturbed until analysis confirms it is Asbestos free.

A sample of the material shall be double packaged in sealed plastic bags, labelled "ASBESTOS - SAMPLE FOR ANALYSIS", placed in a suitable strong mailing envelope and sent to the QatarEnergy-IS Contractor for analysis. The following information shall accompany the sample and shall be copied to HSE:

- Name of worksite
- Date sample taken
- Date sample dispatched for analysis
- Exact description of location from which sample was taken

Note: Warning signage shall be posted on any material that is suspect ACM - as per the sign shown in Attachment 1.

### 4.1.2 ASSESSMENTS

Before carrying out work, which is likely to expose personnel to airborne fibres an adequate risk assessment of exposure shall be made, and a plan of work prepared. The purpose of this is to determine the likely exposure and to specify the steps required to prevent or reduce exposure to the lowest level reasonably practicable.

Written assessments shall be made and shall be reviewed if there is reason to suspect they are no longer valid.

Information on likely airborne fibre concentrations may be known from previous similar work.

In the first instance, an assessment shall be carried out to determine if the intended work is of a Major or Minor nature.

### 4.1.3 PLANS OF WORK

**Major Works**

The licensed specialist Contractor shall carry out an assessment if it is determined that the work is Major. In this case, the method statement shall be developed by the Contractor and shall cover:

- Nature and probable duration of the work
- Location of the work
- Controls to be applied to reduce exposure to Asbestos fibers
- Disposal Procedure for removed ACM.

**Minor Works**

Minor work shall be carried out as per the following:

The requestor shall give good reason for the work requirement.

- A risk assessment of the task **shall be** carried out
- The HSE Manager shall approve the justification and the risk assessment, and
- Work **shall be** carried out under the Permit-to-Work System.

### 4.1.4 PREVENTION OR REDUCTION OF EXPOSURE

ACM that is sound, undamaged and not releasing dust shall not be disturbed. The release of Asbestos dust should be avoided as far as possible.

Remedial actions are:

- Leave the ACM in place without sealing and introduce a stringent management system

- Leave the ACM in place, and effectively manage
- Effectively seal (e.g. encapsulate (paint) or protect by mechanical means – board the material) or
- Remove and dispose of the ACM.

### 4.1.5 MANAGEMENT

It is not normally necessary to seal, enclose or remove ACM, which are sound, undamaged, and not releasing dust. These materials shall be managed as follows:

- A site register noting the areas and condition of the ACM shall be maintained
- Periodic inspections shall be carried out six monthly to ensure that the condition of the material has not changed
- Where necessary minor repairs can be carried out to encapsulate ACM, this work shall be controlled under a risk assessment and the permit to work system
- All personnel that work in the areas where ACM maybe present shall be informed and instructed on minor work.
- The ACM shall be clearly labelled in accordance with Best Industry Practice.

### 4.1.6 SPECIALIST CONTRACTOR

A competent specialized Asbestos Contractor shall undertake all Major Work. This Contractor shall comply with the Best Industry Practice as referred to in this Procedure. They shall make all arrangements for:

- Use and maintenance of control measures
- Provision and cleaning of protective clothing
- Designated areas
- Air monitoring
- Health records and surveillance
- Washing and changing facilities
- Cleaning and clearance certification at end of work

The specialist contractor to remove ACM on a QatarEnergy-IS worksite shall provide a Method Statement, which shall as a minimum contain the following information:

- Name and address of the Asbestos removal contractor
- Name and address of the site to which the method statement relates
- Name of the Supervisor and the appointed Safety Advisor (plus attached C.V.s)
- Arrangements for monitoring the work
- Type of work e.g. removal of boarding – insulation etc.
- Type and quantity of Asbestos
- Results of any analysis
- Probable duration of the work
- The controls to be applied to reduce exposure other than by PPE
- Details of any expected exposure

- Details of the steps to be taken to control the release of Asbestos to the environment
- Arrangements for power supply and clean water and decontamination unit
- Details of the equipment, including PPE, to be used for the protection and decontamination of those carrying out the work.
- Procedures for removal of waste from the work area and the site.
- Procedures for disposal of contaminated water
- Procedures for dealing with emergencies

This checklist is not exhaustive of every issue that may be required for a given task but is intended to assist QatarEnergy-IS management in the approval of Method Statements.

### 4.1.7 ADDITIONAL REQUIREMENTS:

- The specialized Contractor shall submit a report to QatarEnergy which inclusive of identification, sampling result, location and status of specific removal for future reference.
- The specialized Contractor shall declare the site as Asbestos free.
- Local exhaust ventilation, if provided shall be with HEPA filter and dust collection system to prevent spread of asbestos fibre to the environment or to other rooms.
- HVAC system during removal of ACM shall be isolated to prevent spread of asbestos fibre through centralized air conditioning system and contaminate other area.
- The specialized Contractor shall conduct Continuous air monitoring for the work area and personnel exposure monitoring for the asbestos exposure. The monitoring report shall be submitted to QatarEnergy after completion of the project. Reference for the methodology: NMAM Method 7400 (US) and MDHS39/4 Asbestos fibres in air (UK)
- All Activities related to ACM shall be conducted under Permit to Work system. This includes sampling, handling, testing and removal of ACM.
- Area containing ACM during ACM related activity i.e maintenance, sampling & removal access shall be restricted and barricaded

### 4.1.8 SUPERVISION OF THE SPECIALIST CONTRACTOR

The specialist Contractor has responsibilities to ensure that other people who work on the site are not put at risk. QatarEnergy-IS shall execute this responsibility by taking the following steps:

- Ensuring that only contractors that have the necessary license, experience and competence shall be employed
- Ensuring that a suitable and sufficient Method Statement shall be approved by QatarEnergy-IS outlining the safe method of carrying out the work
- Ensuring that Health and safety issues shall be raised at the initial assessment
- Checking that the facilities to be provided by the Contractor are fit for purpose and shall not obstruct accesses / egresses

- Verifying that before work commences the enclosure within which the contractors' personnel will work, shall not permit any escape of Asbestos fibers into the atmosphere
- Ensuring that the enclosure has a viewing panel so that the internal work can be monitored
- Verifying that the Contractor has a maintenance / inspection schedule in place for the enclosure and the air extraction equipment
- Verify that the contractors plan ensures no escape of ACM into the atmosphere during the stripping operation
- Verify that an adequate decontamination Procedure is in place, so that Asbestos fibers etc. are not released from Contractor persons moving through the site
- Verify that the stripped area is clean both visually and supported by air sampling results
- Verify that the removal of the enclosure does not create the release of Asbestos fibers
- Verify that the storage of any destructed Asbestos on site does not create the release of Asbestos fibers, and that all asbestos removed is double bagged and properly sealed.
- Verify that a person with sufficient experience and competence to ensure that all of the above is complied with oversees the work.

### 4.1.9 NOTIFICATION AND COMPLIANCE

The local regulatory body may be required to be notified in writing in advance Legal Department shall advise HSE Department on Regulatory requirements.

### 4.1.10 SAMPLING

All sampling shall be carried under the requirements of OSHA 1910.1001 Appendices A & B, or the equivalent UK requirements, detailed in Control of Asbestos at Work Regulations 2002.

This task shall be controlled through the HSE Department. All sampling is to be requested through the HSE Manager.

### 4.1.11 INFORMATION AND INSTRUCTION

Adequate information and instruction shall be given to personnel who are to undertake Minor Work on ACM. For Major Work, this is the responsibility of the licensed contractor.

This shall include:

- Health hazards of asbestos
- Correct use and maintenance of control measures
- Procedures for reporting defects
- Work methods, hygiene Procedures
- Cleaning of worksite
- Waste disposal arrangements

## 5.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

| Document Title | Document Number | Internal/External Document (if any) | Document Referencing |
|---|---|---|---|
| OSHA General Regulation 1910.1001 “Asbestos”. | OSHA 1910.1001 | External | Sideways |
| Control of Asbestos Regulations 2012 (HSE-UK) | CAR 2012 | External | Sideways |

## 6.0 DEFINITIONS

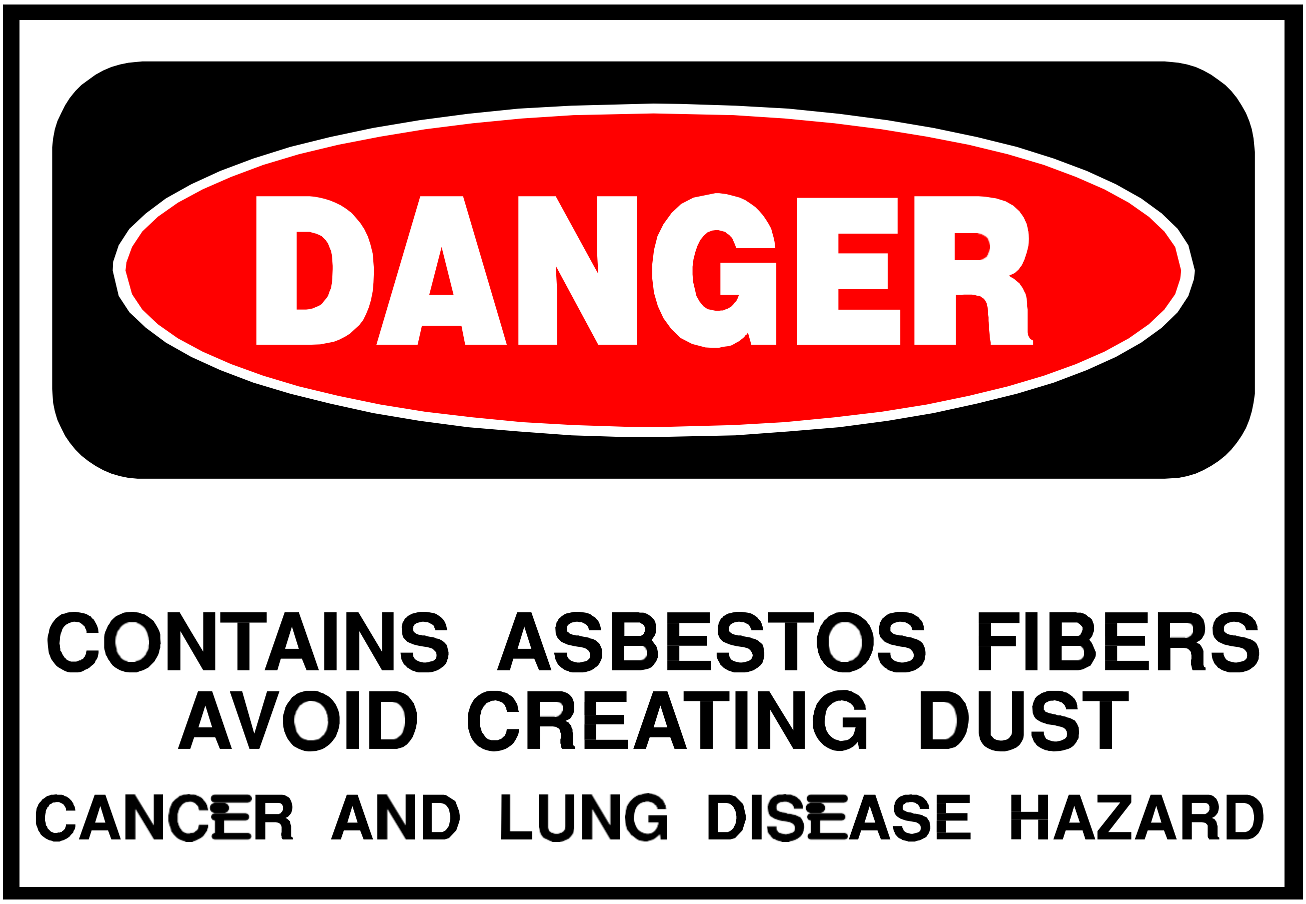
TABLE 2: KEY TERMS DEFINED

| Term | Definition |
|---|---|
| ACM | “Asbestos Containing Material” having more than 1% of Asbestos. |
| Action Level | This relates to the cumulative exposure to Asbestos over a continuous period of 12 weeks. If they are expected to be exceeded during work with ACM then requirements for notification, medical surveillance and designation of work areas are triggered. Action levels for the different forms of Asbestos are given in the UK “Control of Asbestos at Work Regulations 2002” and OSHA General Regulation 1910.1001. |
| Asbestos | Asbestos is a fibrous, physically and chemically inert material extensively used as a fire retardant and insulation material. There are three common types. White (chrysotile), brown (amosite) and blue (crocidolite). All types of asbestos are known to have the potential to cause lung disease and are carcinogenic. |
| Best Industry Practice | Where no regulation exists, best industry practice is the adoption of regulation and Procedure used within that specific industry in another place. |
| Clearance Level | As defined in the UK “Control of Asbestos at Work Regulations 2002” - Areas where Asbestos work has been undertaken should be cleaned such that the level of airborne dust is as low as is reasonably practicable. Clearance monitoring should ensure that Asbestos concentration does not exceed 0.01 fibers per ml of air. This level should not, however, be taken as an acceptable permanent environmental level of Asbestos fiber in air. |
| Control Limit | This defines the levels of Asbestos fiber in inhaled air, which shall not be exceeded. Control Limits averaged over continuous periods of 4 hours and |

| **Term** | **Definition** |
|---|---|
| | 10 minutes are given in the UK Asbestos Regulations and OSHA General Regulation 1910.1001. |
| Major Works | An Action Level or Control Limit is likely to be exceeded, or The duration of the work is longer than one hour or the area of material involved is greater than 2 m2 and in either case, the work involves breaking, cutting, machining, drilling, surface abrasion or any work on a suspended ceiling, i.e. any type of activity, which has the potential for releasing Asbestos fibers into the air.<br>**A licensed specialist Contractor shall carry out all Major Works** |
| Method Statement | A statement from the licensed Contractor detailing how the work is to be carried out safely |
| Minor Works | Minor work is generally that which does not cause the release of Asbestos fibres. Examples are:<br>• Minor electrical or plumbing work<br>• Handling materials in which the Asbestos is well bonded<br>• Small scale work not involving cutting or breaking of ceiling/floor tiles or Asbestos cement<br>  • Fitting of gaskets.<br>  • Painting |

# APPENDIX

## APPENDIX A - ASBESTOS WARNING SIGN



## APPENDIX B – REVISION HISTORY LOG

TABLE 3: REVISION HISTORY LOG **DATE:**

| Item Revised | Revision Description | Page No. |
|---|---|---|
| All | Document updated based<br>- New Management system format<br>- New clause 4.1.7 added based on review feedback | - |
| **Remarks:** | | |

## APPENDIX C – LIST OF ABBREVIATIONS

TABLE 4: LIST OF ABBREVIATIONS

| Term | Definition |
|---|---|
| ACM | Asbestos Containing Material |
| QatarEnergy-IS | Idd El-Shargi Asset of QatarEnergy. |